# NLP Interview Questions

## Section 1: NLP Fundamentals

1. What is Natural Language Processing (NLP), and what are its main applications?
2. What is the difference between Natural Language Processing, Natural Language Understanding (NLU) and Natural Language Generation (NLG)?
3. How does NLP differ from traditional text processing?
4. What are the major challenges in processing human language?
5. What is the difference between structured, semi-structured and unstructured data?
6. What are lexical, syntactic, semantic and pragmatic analysis?
7. What is ambiguity in natural language? Explain lexical and syntactic ambiguity with examples.
8. What is a corpus in NLP, and how is it used to develop language models?
9. What is the difference between rule-based NLP, statistical NLP and neural NLP?
10. What are the common real-world applications of NLP in healthcare, finance, education and customer service?

## Section 2: Text Preprocessing and Normalisation

11. What is text preprocessing, and why is it important in NLP?
12. What is text normalisation? Give examples of normalising informal text.
13. What is tokenisation, and what are the different types of tokenisation?
14. What is the difference between sentence tokenisation and word tokenisation?
15. What is the difference between stemming and lemmatisation?
16. What are stop words, and when should they be removed?
17. What is case folding, and when might converting text to lowercase be inappropriate?
18. How would you handle punctuation, numbers, URLs and special characters during preprocessing?
19. What is the difference between whitespace tokenisation, subword tokenisation and character tokenisation?
20. How would you preprocess noisy social media text containing emojis, hashtags, abbreviations and spelling mistakes?

## Section 3: Regular Expressions and Text Processing

21. What are regular expressions, and how are they useful in NLP?
22. What is the difference between `re.match()`, `re.search()` and `re.findall()` in Python?

23. How would you extract email addresses and telephone numbers from text using regular expressions?
24. How would you remove HTML tags from a text document?
25. How would you identify and extract hashtags, mentions and URLs from social media posts?
26. What is the difference between greedy and non-greedy matching in regular expressions?
27. How would you remove duplicate whitespace and standardise line breaks in text?
28. What are Unicode and UTF-8, and why do they matter in multilingual NLP?
29. How would you detect and correct common spelling errors in text?
30. What are the limitations of using regular expressions as the primary method for understanding natural language?

## Section 4: Linguistic Analysis

31. What is morphology, and how does it relate to NLP?
32. What are morphemes, and what is the difference between free and bound morphemes?
33. What is Part-of-Speech (POS) tagging?
34. What are nouns, verbs, adjectives, adverbs and conjunctions in the context of POS tagging?
35. What is syntactic parsing, and how does it differ from POS tagging?
36. What is a constituency parse tree?
37. What is dependency parsing, and how does it represent relationships between words?
38. What is Named Entity Recognition (NER), and which types of entities can it identify?
39. What is coreference resolution, and why is it important for understanding documents?
40. What is word sense disambiguation? Explain how a word can have different meanings depending on context.

## Section 5: Text Representation and Vectorisation

41. Why must text be converted into numerical representations before being processed by most ML algorithms?
42. What is a Bag-of-Words (BoW) representation?
43. What are the advantages and limitations of Bag-of-Words?
44. What is an n-gram, and how do unigram, bigram and trigram representations differ?
45. What is a document-term matrix?
46. What is Term Frequency–Inverse Document Frequency (TF-IDF)?
47. How does TF-IDF differ from simple term frequency?
48. What is the difference between CountVectorizer and TfidfVectorizer in Scikit-learn?
49. What is vocabulary size, and how can a very large vocabulary affect an NLP model?
50. How would you represent a collection of text documents for a sentiment classification model using Python?

## Section 6: Word Embeddings

51. What is a word embedding, and how does it differ from one-hot encoding?
52. What does it mean for words to have similar representations in an embedding space?
53. What is Word2Vec, and how does it learn word representations?
54. What is the difference between Continuous Bag-of-Words (CBOW) and Skip-gram?
55. What is GloVe, and how does it use global word co-occurrence statistics?
56. What is FastText, and how does it use character n-grams?
57. What are the limitations of static word embeddings?
58. What is the difference between static and contextual word embeddings?
59. How can word embeddings represent semantic relationships between words?
60. How would you choose between TF-IDF, Word2Vec, FastText and pretrained contextual embeddings for a text classification task?

## Section 7: Traditional Machine Learning for NLP

61. How can Naive Bayes be used for text classification?
62. Why is Multinomial Naive Bayes commonly used for document classification?
63. How can Logistic Regression be applied to sentiment analysis?
64. How can Support Vector Machines classify text documents?
65. What are the advantages of linear models for high-dimensional, sparse text data?
66. What is sentiment analysis, and what are its different levels?
67. What is the difference between binary, multiclass and multilabel text classification?
68. How would you build a spam detection model using TF-IDF and Logistic Regression?
69. What evaluation metrics would you use for an imbalanced text classification dataset?
70. When would you prefer a traditional ML model over a deep neural network for an NLP problem?

## Section 8: Sequence Models and Recurrent Neural Networks

71. Why is word order important in NLP?
72. What is a sequence model, and how does it differ from a Bag-of-Words model?
73. What is a Recurrent Neural Network (RNN)?
74. What is the hidden state of an RNN, and how does it carry information across time steps?
75. What is Backpropagation Through Time (BPTT)?
76. What are the vanishing and exploding gradient problems in recurrent networks?
77. What is a Long Short-Term Memory (LSTM) network?
78. What are the forget, input and output gates in an LSTM?
79. What is a Gated Recurrent Unit (GRU), and how does it differ from LSTM?

80. How would you build a sequence classification model for detecting sentiment in customer reviews?

## Section 9: Attention and Transformer Architecture

81. What is the attention mechanism, and why was it introduced?
82. What is self-attention?
83. What are query, key and value vectors in an attention mechanism?
84. How does scaled dot-product attention work?
85. What is multi-head attention, and what advantages does it provide?
86. What is positional encoding, and why do Transformers need positional information?
87. What is the difference between an encoder and a decoder in a Transformer?
88. What is causal masking, and why is it used in autoregressive language models?
89. Why can standard self-attention be expensive for long sequences?
90. What are the main differences between RNN-based models and Transformer-based models?

## Section 10: Pretrained Language Models

91. What is a pretrained language model, and how is it trained?
92. What is BERT, and how does its bidirectional context representation work?
93. What is GPT, and how does autoregressive language modelling work?
94. What is the difference between BERT and GPT in architecture and training objectives?
95. What is a masked language modelling objective?
96. What is the difference between pretraining and fine-tuning?
97. What is transfer learning in NLP, and why is it useful when labelled data is limited?
98. What is a tokenizer in a pretrained Transformer model?
99. What are WordPiece, Byte-Pair Encoding (BPE) and SentencePiece?
100. How would you fine-tune a pretrained Transformer for sentiment classification?

## Section 11: Advanced NLP Tasks

101. What is machine translation, and how do sequence-to-sequence models perform it?
102. What is text summarisation, and how do extractive and abstractive summarisation differ?
103. What is question answering in NLP, and what are extractive and generative approaches?
104. What is text generation, and how is it different from text classification?
105. What is semantic textual similarity, and how can it be measured?
106. What is sentence embedding, and how does it differ from word embedding?
107. What is semantic search, and how does it differ from keyword-based search?
108. What is intent classification, and how is it used in chatbots?

109. What is slot filling, and how does it complement intent classification?
110. How would you build a multilingual NLP application that supports English and Malayalam?

## Section 12: Large Language Models and Generative NLP

111. What is a Large Language Model (LLM), and what makes it different from traditional NLP models?
112. What is next-token prediction, and how does it train an autoregressive language model?
113. What are parameters and context windows in an LLM?
114. What is the difference between temperature, top-k sampling and top-p sampling?
115. What are hallucinations in LLMs, and why do they occur?
116. What is prompt engineering, and how can prompt structure affect model outputs?
117. What is instruction tuning, and how does it differ from general pretraining?
118. What is Reinforcement Learning from Human Feedback (RLHF)?
119. What is quantisation in LLMs, and how can it reduce memory requirements?
120. What is the difference between full fine-tuning, parameter-efficient fine-tuning and inference-time prompting?

## Section 13: Retrieval-Augmented Generation and NLP Applications

121. What is Retrieval-Augmented Generation (RAG), and why is it useful?
122. What are the main stages of a RAG pipeline?
123. What is text chunking, and how do chunk size and overlap affect retrieval quality?
124. What are embeddings and vector databases, and how do they support semantic retrieval?
125. What is cosine similarity, and how is it used to compare text embeddings?
126. What is the difference between dense retrieval, sparse retrieval and hybrid search?
127. What is a reranker, and how does it improve retrieval results?
128. How would you evaluate retrieval quality separately from the quality of generated answers?
129. How would you reduce hallucinations in a RAG-based question-answering system?
130. How would you build a document question-answering system using Python, a pretrained embedding model and a vector database?

## Section 14: NLP Evaluation, Data Quality and Deployment

131. What are accuracy, precision, recall and F1-score in text classification?
132. What is the difference between micro, macro and weighted F1-score?
133. What are BLEU and ROUGE, and what tasks are they commonly used to evaluate?

134. What is perplexity in language modelling, and what does it indicate?
135. Why might a lower perplexity not necessarily mean that a model produces better answers?
136. What is data leakage in NLP, and how can it occur during text preprocessing or dataset splitting?
137. What is distribution shift in NLP, and how can changing language patterns affect model performance?
138. How would you deploy an NLP model through a REST API using FastAPI or Flask?
139. How would you monitor latency, throughput, prediction quality and data drift in a production NLP system?
140. What privacy, bias and fairness issues should be considered when deploying NLP applications?